



**SWAMINARAYAN  
UNIVERSITY**  
THE RESERVOIR OF ALL KNOWLEDGE

Faculty of Engineering  
Department of Computer Engineering

| Course Code | DE203101                           | Semester              | 1                                                   |    |                   |    |             |               |  |
|-------------|------------------------------------|-----------------------|-----------------------------------------------------|----|-------------------|----|-------------|---------------|--|
|             |                                    |                       | Teaching & Learning Scheme (In Hours Per Semester ) |    |                   |    |             |               |  |
| Category    | PROFESSIONAL CORE                  | Classroom Instruction |                                                     |    | Extended Learning |    | Total Hours | Total Credits |  |
|             |                                    | L                     | T                                                   | P  | TW                | SL | HR          | C             |  |
| Course Name | COMPUTER APPLICATIONS & WEB DESIGN | 0                     | 15                                                  | 60 | 5                 | 10 | 90          | 3             |  |

### Course Objectives

- To understand the fundamentals of computers, operating systems, and internet technologies.
- To develop proficiency in using productivity tools such as MS Word, Excel, and PowerPoint.
- To learn the concepts and techniques of static web page design using HTML.
- To apply CSS for styling and formatting web pages effectively.
- To develop interactive web pages using JavaScript.

### Prerequisite

- Basic understanding of computer operations.
- Familiarity with keyboard and file management.
- Basic knowledge of internet usage.

|                   |                                 |                        |
|-------------------|---------------------------------|------------------------|
| <b>Syllabus –</b> | <b>Total Contact Hours - 15</b> | <b>No of Credits-3</b> |
|-------------------|---------------------------------|------------------------|

### UNIT-I Basics of Computer and Internet

**3 Hrs**

Introduction to Computer, Generations of Computers, Operating System Concepts, Functions of Operating System, Windows Basics, File and Folder Management, Control Panel, Taskbar, Software Installation, Internet Fundamentals, Browser, Website, Webpage, URL, WWW, E-mail, Gmail Features, Google Drive, Google Forms.

### UNIT-II Microsoft Office Tools

**3 Hrs**

MS Word: Text Editing, Formatting, Tables, Images, Mail Merge.  
MS Excel: Formulas, Functions, Sorting, Filtering, Conditional Formatting, Charts, Pivot Tables.  
MS PowerPoint: Slide Creation, Themes, Layouts, Slide Master, Slide Show, Custom Presentations.

### UNIT-III Basics of HTML

**3 Hrs**

Introduction to HTML, Tags and Attributes, Formatting Tags, Image Tags, Hyperlinks, Tables, Captions, Table Structures.

#### **UNIT-IV Advanced HTML and CSS**

**3 Hrs**

Lists, Semantic Elements, Media Tags, HTML Forms, CSS Types, CSS Selectors, Div Elements, Styling Web Pages, Class and ID Selectors.

#### **UNIT-V JavaScript**

**3 Hrs**

JavaScript Basics, Variables, Data Types, Operators, Built-in Functions, Conditional Statements, Loops, User Defined Functions, DOM, Form Handling, HTML Events.

#### **Term Work Module**

**5 hours**

##### **Objective:**

1. To understand computer applications and web technologies.
2. To develop skills in MS Office tools and internet applications.
3. To design static web pages using HTML and CSS.
4. To create interactive web pages using JavaScript.

#### **Term Work 1: Computer Applications Assignment (2 Hours) (C01 & C02)**

Students will:

- Create and manage Gmail accounts and Google Drive resources.
- Prepare professional documents using MS Word.
- Perform data analysis using Excel functions and charts.
- Create presentations using PowerPoint.

#### **Term Work 2: Web Design Practical Assignment (2 Hours) (C03 & C04)**

Students will:

- Design HTML web pages using formatting and media tags.
- Develop forms and tables.
- Apply CSS for page styling.
- Create responsive and structured web layouts.

#### **Term Work 3: Mini Project / Viva (1 Hours) (C01 to C05)**

Students will:

- Develop a small website.
- Present project implementation.
- Submit project report and documentation.
- Demonstrate practical understanding through viva.

#### **Self-Learning Module**

**10 hours**

##### **Objective:**

To supplement classroom learning through self-study using online resources related to office automation, web design, HTML, CSS, and JavaScript.

##### **Recommended Learning**

- Advanced MS Office features.

- HTML5 and CSS3.
- JavaScript fundamentals.
- Google Workspace applications.
- Responsive web design concepts.

### **Recommended Courses:**

1. SWAYAM – Web Design Courses
2. NPTEL – Introduction to Internet Technologies
3. Coursera – HTML, CSS and JavaScript for Web Developers
4. W3Schools HTML/CSS/JavaScript Tutorials Link:

## **Teaching–Learning Process / Pedagogy**

### **Computer Applications & Web Design**

#### **1. Lecture-Based Teaching–Learning**

Used for explaining fundamental concepts of computers, operating systems, internet technologies, office automation tools, HTML, CSS, and JavaScript. Faculty members deliver structured lectures supported by presentations, demonstrations, and real-world examples to build conceptual understanding.

#### **2. Demonstration-Based Learning**

Faculty demonstrate practical usage of computer applications such as MS Word, Excel, PowerPoint, Google Drive, and Google Forms. Live demonstrations are also used to illustrate webpage creation, formatting, styling, and scripting techniques.

#### **3. Laboratory-Based Learning**

Students perform hands-on practical exercises in computer laboratories. They gain experience in document creation, spreadsheet analysis, presentation design, webpage development, CSS styling, and JavaScript programming through guided laboratory sessions.

#### **4. Activity-Based Learning**

Students actively participate in classroom and laboratory activities such as creating online forms, designing web pages, preparing reports, creating presentations, and developing small web-based applications. These activities help reinforce theoretical concepts through practical implementation.

#### **5. Project-Based Learning**

Students undertake mini-projects involving the development of websites, online registration systems, departmental web portals, portfolio websites, or business information websites. Project work encourages creativity, problem-solving skills, and practical application of web technologies.

#### **6. Collaborative Learning and Group Activities**

Students work in teams to design web pages, prepare presentations, conduct peer reviews, and complete assignments. Group discussions and collaborative projects enhance teamwork, communication skills, and knowledge sharing.

## **7. Inquiry-Based Learning**

Students explore different web technologies, internet tools, and software applications through guided investigations. They are encouraged to identify problems, research solutions, and experiment with various web development techniques independently.

## **8. Problem-Solving Based Learning**

Students solve practical problems related to document formatting, spreadsheet calculations, webpage design, CSS implementation, form validation, and JavaScript programming. Problem-solving activities improve analytical thinking and logical reasoning skills.

## **9. Technology-Enhanced Learning**

Modern ICT tools, learning management systems, online tutorials, coding platforms, multimedia resources, and web-based learning materials are integrated into the teaching process. Students utilize online resources such as W3Schools, SWAYAM, NPTEL, Coursera, and Google Workspace applications.

### **Learning Resources**

#### **Reference Books -**

- Fundamentals of Computers – Rajaraman V.
- MS Office for Dummies – Wallace Wang
- Sams Teach Yourself Internet and Web Basics
- HTML5 Black Book
- HTML & CSS: The Complete Reference
- JavaScript: The Complete Reference

#### **Tutorials-**

**60 hours**

1. Gmail account creation and management.
2. Google Forms creation.
3. MS Word document preparation.
4. Mail Merge implementation.
5. Excel formulas and chart creation.
6. PowerPoint presentation design.
7. HTML formatting tags.
8. Hyperlinks and image insertion.
9. HTML tables.
10. Lists and semantic tags.
11. HTML forms.
12. CSS implementation.
13. CSS profile page design.
14. JavaScript programs.

#### **List of Major Equipment –**

- Computer Systems with Internet Facility
- MS Office Suite
- Web Browsers

- HTML Editors / IDEs
- JavaScript Enabled Environment

#### Course Outcomes-

| CO No | Course Outcome Description                                                      | RBT Level       |
|-------|---------------------------------------------------------------------------------|-----------------|
| CO1   | Understand computer fundamentals, operating systems, and internet technologies. | Understand (C2) |
| CO2   | Use MS Office tools for documentation, data analysis, and presentations.        | Apply (C3)      |
| CO3   | Design static web pages using HTML elements and tags.                           | Apply (C3)      |
| CO4   | Apply CSS for styling and formatting web pages.                                 | Apply (C3)      |
| CO5   | Develop interactive web pages using JavaScript and DOM concepts.                | Apply (C3)      |

#### Mapping With Program Outcomes

| CO \ PO/PSO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1         | 3   | 2   | 1   | 1   | 2   | 0   | 1   | 1   | 1   | 1    | 0    | 2    | 2    | 2    |
| CO2         | 3   | 2   | 2   | 1   | 3   | 0   | 1   | 1   | 1   | 1    | 0    | 2    | 3    | 2    |
| CO3         | 2   | 3   | 3   | 2   | 3   | 0   | 1   | 1   | 1   | 1    | 0    | 2    | 3    | 3    |
| CO4         | 2   | 3   | 3   | 2   | 3   | 0   | 1   | 1   | 2   | 1    | 0    | 2    | 3    | 3    |
| CO5         | 2   | 3   | 3   | 2   | 3   | 0   | 1   | 1   | 2   | 2    | 0    | 3    | 3    | 3    |

#### CO PO Mapping Justification for COMPUTER APPLICATIONS & WEB DESIGN

##### PO1 (Engineering Knowledge):

The course provides fundamental knowledge of automata, formal languages, computability, and complexity theory.

##### PO2 (Problem Analysis):

Students analyze computational problems and language recognition using mathematical and logical methods.

##### PO3 (Design/Development of Solutions):

The course supports development of formal computational models and algorithmic problem-solving techniques.

##### PO4 (Conduct Investigation of Complex Problems):

Students investigate computational limitations and complexity classes through analytical approaches.

##### PO5 (Modern Tool Usage):

Students use automata simulators and computational tools for language and machine analysis.

##### PO6 (The Engineer and Society):

Not applicable limited focus on societal and legal issues.

**PO7 (Environment and Sustainability):**

Basic computational optimization concepts support efficient computing systems.

**PO8 (Ethics):**

Not explicitly covered in this course.

**PO9 (Individual and Team Work):**

Students perform assignments and presentations individually and in groups.

**PO10 (Communication):**

Students prepare technical reports and presentations on computational concepts.

**PO11 (Project Management and Finance):**

Not applicable management and finance concepts are not included.

**PO12 (Life-long Learning):**

The course develops theoretical foundations required for advanced studies and research.

**Mapping With Sustainable Development Goals**

|                                                       |                     | SDG Mapping for – DE203101- COMPUTER APPLICATIONS & WEB DESIGN              |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------|
| SDG No. and Title                                                                                                                       | Relevance to Course | Justification                                                               |
| <b>4</b> QUALITY EDUCATION<br>                       | ✓                   | Digital literacy and web design skills enhance quality education.           |
| <b>8</b> DECENT WORK AND ECONOMIC GROWTH<br>         | ✓                   | Computer application skills improve employability and productivity.         |
| <b>9</b> INDUSTRY, INNOVATION AND INFRASTRUCTURE<br> | ✓                   | Web technologies support digital innovation and infrastructure development. |

|                                                                                                                                   |                                                                                   |                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>10</b> REDUCED INEQUALITIES<br>               |  | Digital access and web technologies improve information accessibility. |
| <b>11</b> SUSTAINABLE CITIES AND COMMUNITIES<br> |  | Web-based information systems support smart community services.        |
| <b>17</b> PARTNERSHIPS FOR THE GOALS<br>         |  | Collaborative projects encourage teamwork and knowledge sharing.       |